WHAT IT IS

ToneTunner is an Android voice input app that runs local Whisper ONNX inference for speech transcription and optional translation to English. It can run as a standalone app, an Input Method Editor (IME), and an Android RecognitionService endpoint.

WHO IT IS FOR

Primary persona: Android users who want offline multilingual speech-to-text and voice input integration in keyboard and system voice-input flows.

WHAT IT DOES

- Records microphone audio at 16 kHz mono PCM with optional WebRTC VAD auto-stop.
- Runs on-device Whisper inference through ONNX Runtime sessions.
- Provides standalone transcription UI with language selector and copy action.
- Supports optional translation mode and optional Text-to-Speech playback.
- Provides IME voice typing and commits recognized text to active input fields.
- Supports speech intent entry point via RECOGNIZE_SPEECH activity.
- Supports system-wide voice input via Android RecognitionService.

KEY ENTRY POINTS

- MainActivity: standalone transcription and translation UI.
- WhisperInputMethodService: IME keyboard voice input.
- WhisperRecognitionService: system speech service integration.
- WhisperRecognizeActivity: popup recognizer for speech intents.
- SetupActivity: model download link and ZIP import workflow.

NOTES

- Inference appears local/on-device in this repo.
- App requests RECORD_AUDIO and related foreground service permissions.